

Unit-1

- Define Machine Learning and explain its scope.
- Differentiate between PCA and Linear Discriminant Analysis (LDA).
- What is Feature Engineering? Explain its importance in Machine Learning.
- Explain parametric and non-parametric models in Machine Learning.
- Differentiate between Artificial Intelligence, Machine Learning, and Data Science with suitable examples.
- Explain Geometric, Probabilistic, and Logical models of Machine Learning. Give one suitable example of each.
- Explain the following types of Machine Learning with suitable examples:
 - Supervised Learning
 - Unsupervised Learning
 - Semi-supervised Learning
 - Reinforcement Learning
- Explain the traditional programming paradigm and Machine Learning paradigm with a suitable example.
- Define regression and explain the **need for regression** with a suitable real-life example.

Unit- 2

- Differentiate between regression and correlation.
- Explain Univariate and Multivariate Regression with suitable examples.
- Differentiate between Simple and Multiple Regression with suitable examples.
- Explain Linear and Nonlinear Regression with suitable examples.
- Explain the Bias–Variance Tradeoff.
- Differentiate between Overfitting and Underfitting with suitable examples.
- Explain Simple Linear Regression and Multiple Linear Regression with their mathematical equations and suitable examples.
- Explain Polynomial Regression with a suitable example.
- Explain Decision Tree Regression and Random Forest Regression. Compare their working principles.
- Explain Support Vector Regression (SVR) and mention its suitable application.
- Explain Ridge Regression (L2) and Lasso Regression (L1).
- Compare Ridge and Lasso Regression based on regularization and their effect on regression coefficients.
- Explain Mean Squared Error (MSE) and Mean Absolute Error (MAE) as regression evaluation metrics.
- Explain Root Mean Squared Error (RMSE) and R-squared (R^2) as regression evaluation metrics.
- Compare MSE, MAE, RMSE and R^2 . Identify which metric is more sensitive to large errors and justify your answer.